GROUP #3: Honkshoe

Element D: Design Concept Generation, Analysis, and Selection

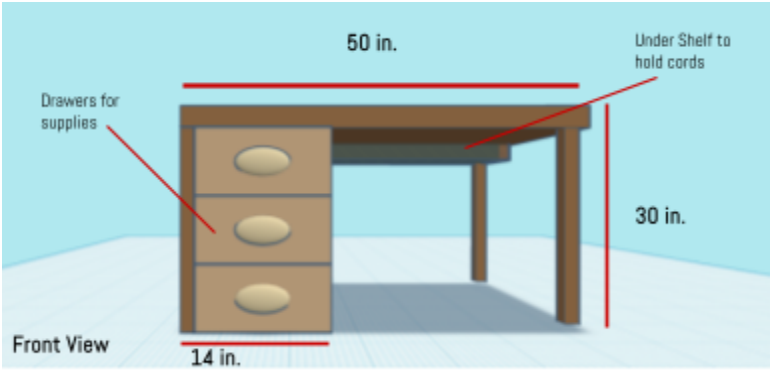
Design Idea Generation (D1)

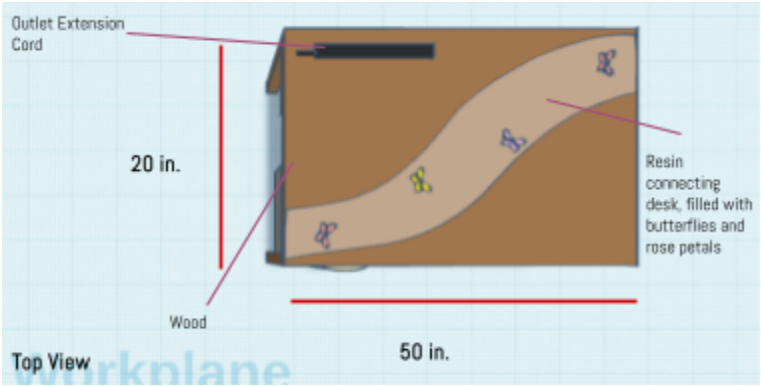
This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

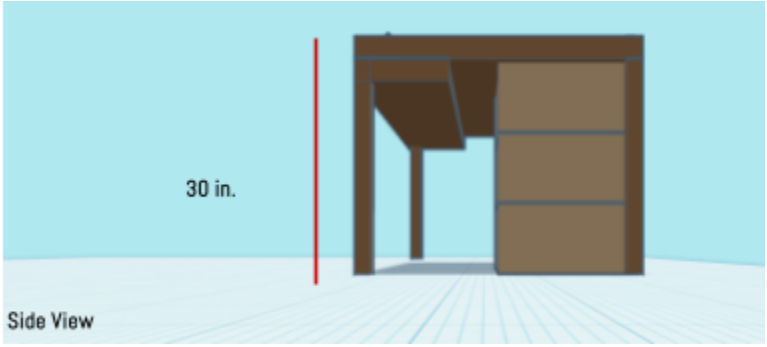
Design Idea #1 | V.T

Description: This is a 3D model of our computer desk for Ms. A.V . The main part of the desk, the surface area measuring 50 by 20 inches, contains two features specifically requested by our stakeholder. The first is a resin 'river', a piece of resin made to connect two parts of wood for the flat area of the desk. Inside the resin contains butterflies and rose petals for aesthetic purposes. The second feature on the surface area of the desk is the outlet extension cords. Since this desk is specifically a computer desk, the stakeholder needs an area to connect and her multiple devices to in order to power the. The cord of the outlet will lead to the undershelf, where it can also alternatively be placed. This allows for the stakeholder to adjust to her preference. Another feature of this desk is the drawers, which are 14x20 inches. Attached to the 2x2 legs; the drawers provide an area for the stakeholder to store necessary documents and materials. This, like the majority of the desk, will be made of wood.

Drawing that includes distinguishing information such as scale, dimensions, and materials:





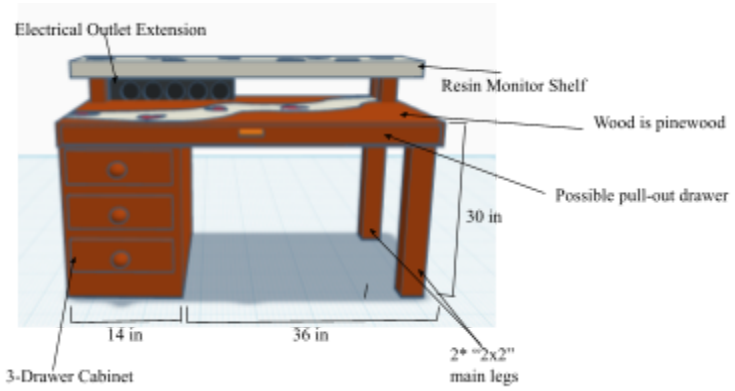


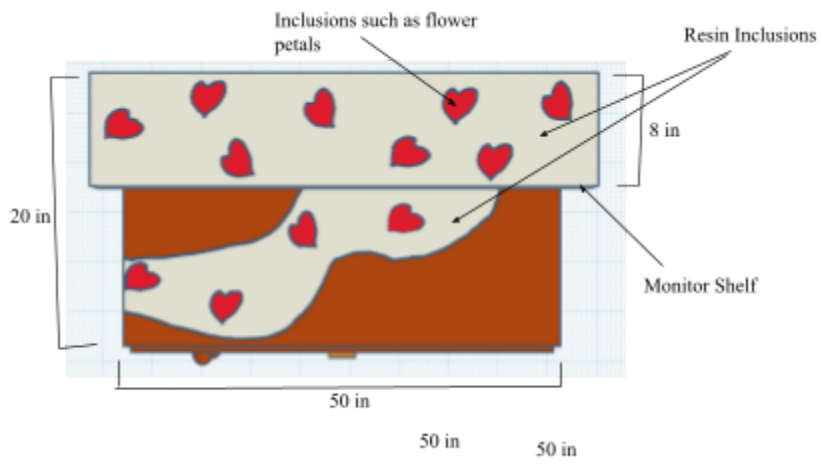
Design Idea #2 | K.Z

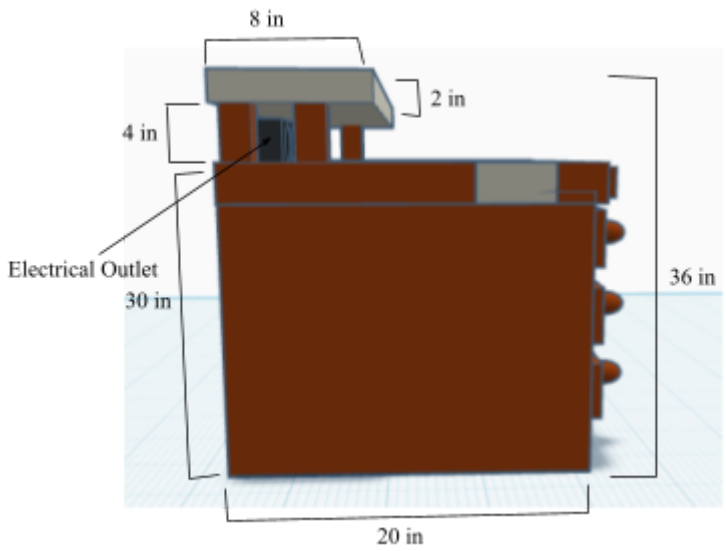
Date: March 10th, 2025

Description: Below is a desk that Ms. A.V would possibly be content with that would meet her needs and wants. The overall height of the desk would be a total of 36 inches, with its depth being 20 inches, and the width being 50 inches. The height would be split into three parts, the main desk, the legs of the monitor shelf, and the thickness of the monitor shelf itself. The majority of the desk would be made out of pinewood, with it having a resin lake, and the monitor shelf being mostly made of resin. The shelf would have a depth of 8 inches, and the same width as the main desk. The desk itself would have two 2x2 legs and a 3-drawer cabinet as its support. It also includes a 5-outlet electrical extension cord that allows for easy access to power for any electronic needed. The resin, as requested by Ms. A.V , has inclusions such as flower petals and butterflies. It would be a translucent-like color as the base. The desk would possibly also have a long, singular drawer right at the base of the tabletop itself for storage or a possible keyboard and mouse.

Drawing that includes distinguishing information such as scale, dimensions, and materials:







Design Idea #3 | C.G

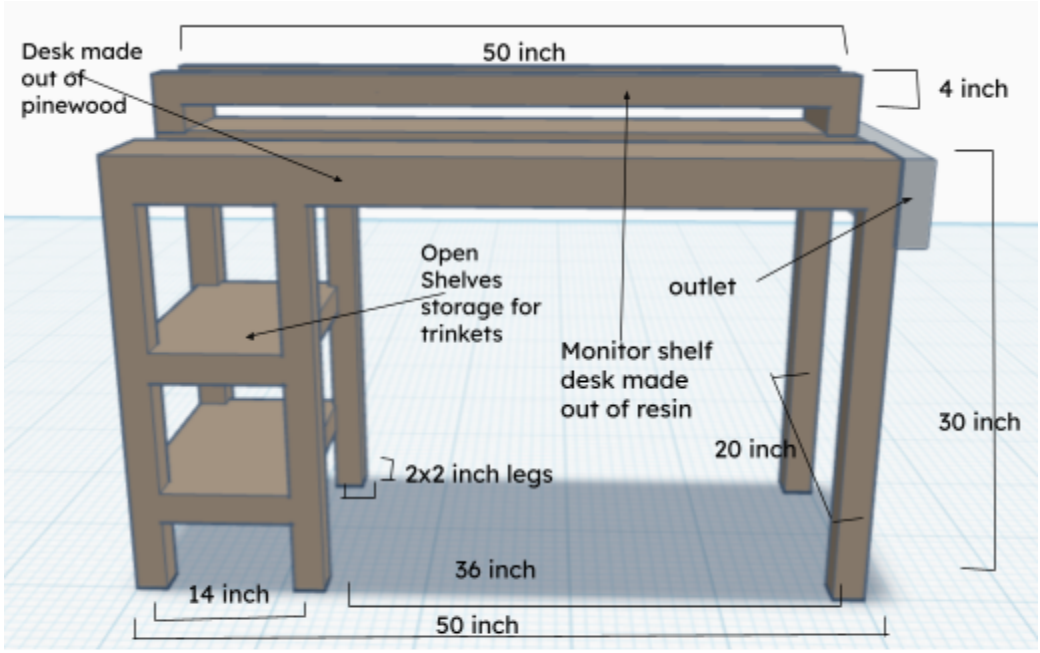
Date: March 10th, 2025

Description: This is a 3D model of our computer standard desk that includes a main surface area desk made out of pinewood and not only has a main surface desk area, but also includes a monitor shelf/2nd level of the desk specifically designed to place the stakeholder's monitor. This desk is a 50 x 30 desk that comes with a 50 inch monitor shelf that is 4 inches above the main desk surface and the main desk surface and monitor shelf includes white epoxy resin art with inclusions of rose petals.The main desk surface is a combination of this resin art with pine wood and the monitor shelf is made completely of epoxy resin with the inclusions. There is also an outlet at the side which can be used to plug in your monitor connections, charges, etc.. In addition, at the bottom there are two open shelves with 14 inches of length and 20 inches of depth where these shelves can be used for trinkets and to place journals, store office items, etc.. Furthermore, the 3D model below shows a design of this standard office/computer desk that is great ergonomics and matches with this description:

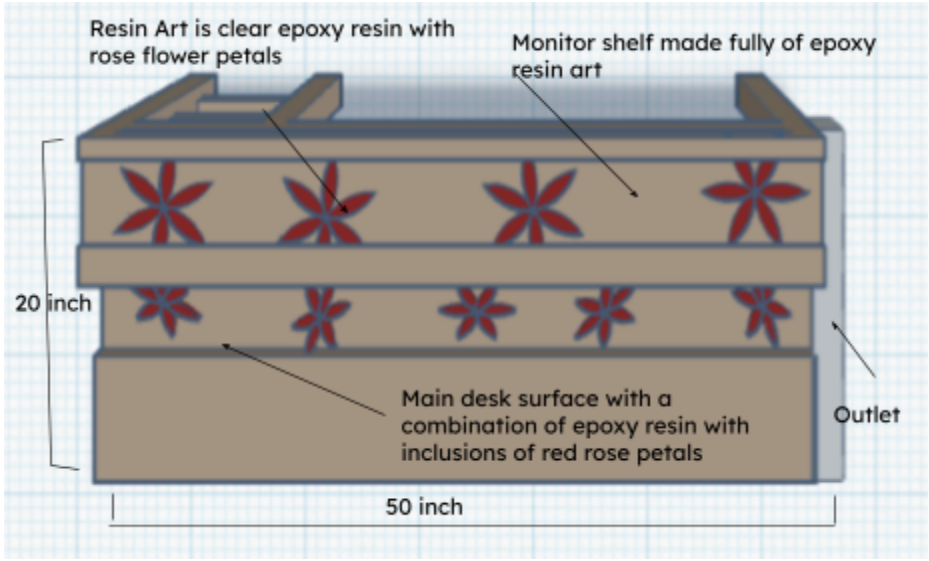
Drawing that includes distinguishing information such as scale, dimensions, and materials:

- Desk made out of **pinewood**
- **Resin Art (Epoxy Resin with inclusions, red rose petals)**

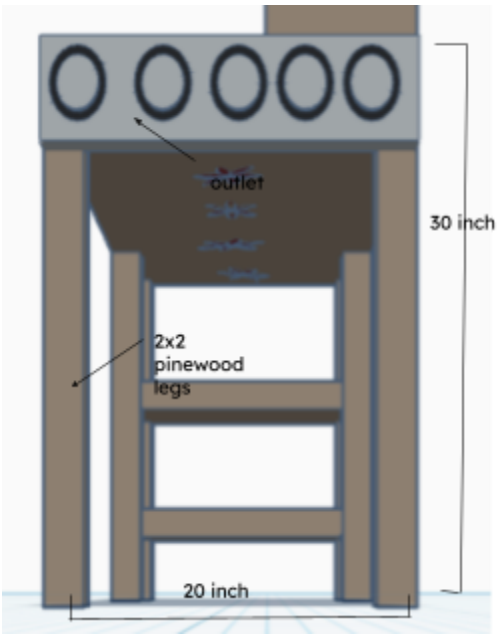
Front view: | ⅓ scale ratio

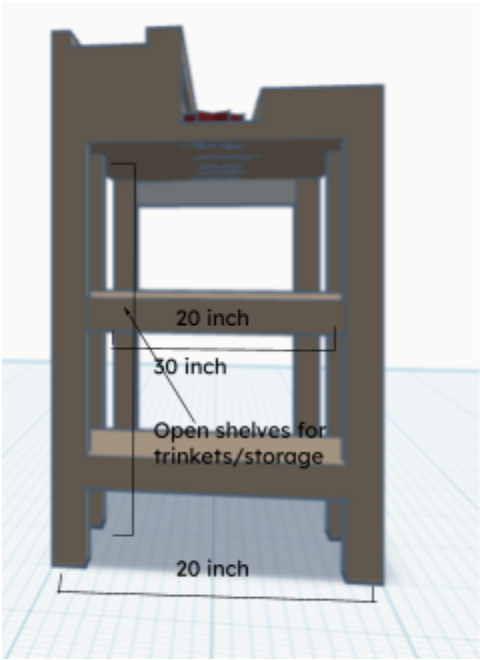


Top view:



Side view:



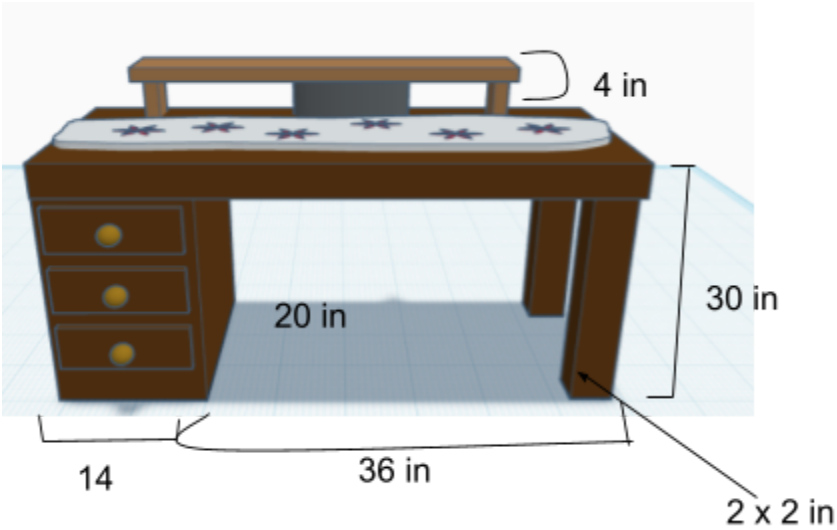


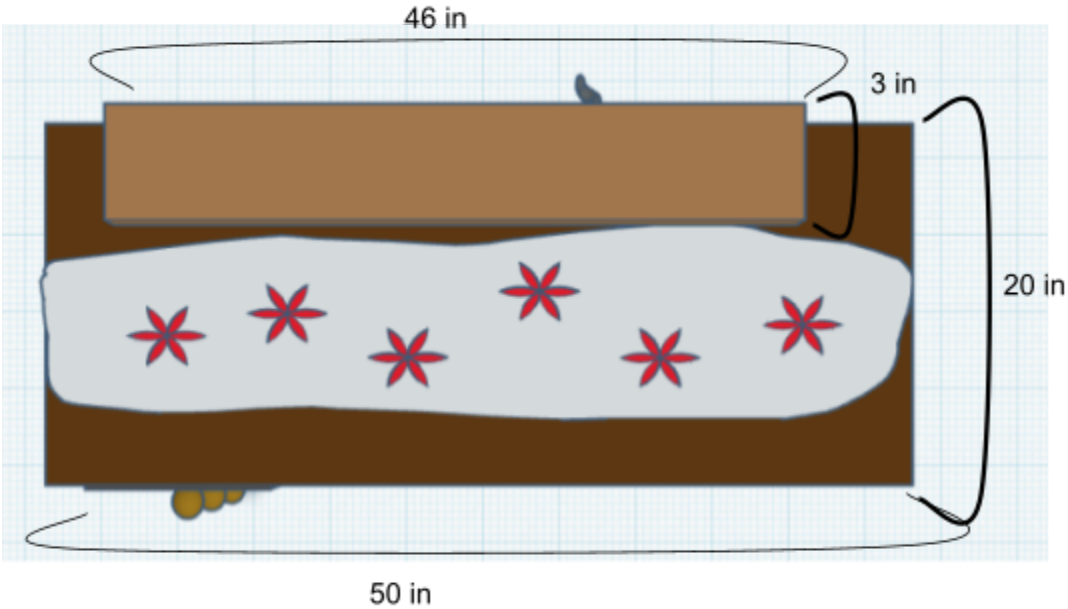
Design Idea #4 | O.B

Date: March 10th, 2025

Description: This is a 3D model of a computer desk that Ms. A.V would like. The desktop is 50 x 20 inches so that Ms. A.V would have a lot of room to work. The main desk is made up of wood and resin to have a resin driver. The river will include red rose flowers and butterflies. There is also a monitor/trinket stand that is 4 inches off the original height of the desk which is 30 inches. The monitor/trinket holder is 46 inches long. The cabinets that are attached to the desk are 20 inches in depth and 14 inches in width. There will be three

drawers. The reason that the trinket shelf is 4 inches off the original desk is because our desk is a computer desk and our stakeholder would like a place to attach cords. So with that extra space we can attach an outlet station. This desk will be made out of epoxy resin, cherry wood, fake flowers, and fake butterflies.





Weighted Decision Matrix - Pugh Matrix(D2)

Design Criteria	Weight	Victoria T.	Keli Z.	Olivia B.	Caylee G.
Stability/Structural Integrity	0.30	4	3	3	4
Affordability	0.15	2	2	1	3
Resin Art	0.30	2	3	2	2
Outlet	0.5	3	2	3	2
Trinket Shelf	0.10	0	3	2	3
Pastel Colors	0.05	0	2	-1	1
Score		1.83	2.50	1.83	2.60

Design that will be used for prototype: **Caylee's Design**

After completing this pugh matrix chart, we finalized that the design that we will use as a reference and guide for our prototype of our standard computer desk is Caylee's design. T.Bsed on these scores and analyzing them we came to this conclusion that we use Caylee's design due to it meeting all the requirements and design criteria that we have for our community project. Not only does it meet the design criteria for our prototype, but also meets the needs and wants for our stakeholder. Caylee design has the structural integrity where the legs will be 2x2 which is a really good size for a leg desk for durability and will be able to withstand any weight. Also, the desk is made out of pinewood which is a good and durable quality of wood, but also affordable which helps us as the builders to have a less burden when it comes to cost and lets us breathe a little with the budget we decide (at least $200). In addition, it meets the requirement of having trinket shelves where the desk has open shelves under the main surface desk area where the stakeholder can place their trinkets or other stuff. Furthermore, the design expresses the resin art and has the outlet included at the side of the desk resulting in the design meeting a lot of the requirements. However, a trade-off was made in affordability, as Caylee's design was not the cheapest, but it excelled in structural integrity and trinket shelf functionality. Overall, it is a good design where it meets most of the

requirements and design criteria and that's why Caylee's design was chosen to be our prototype.

Gantt Chart

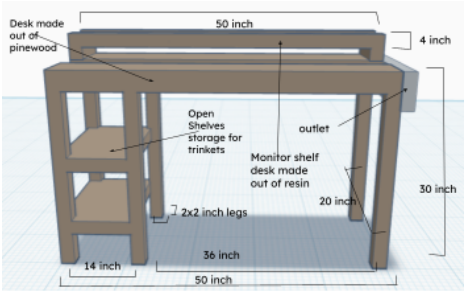
Task	Progress / Completion	Materials	Start Date	End Date	Status
Conceptual Design	----------	----------	----------	----------	
Ranking Design requirements - *Determining design requirements of project and ranking it's importance*	100%	Google Docs	3/5/25	3/7/25	Completed
Sketching Initial Design Ideas - *Using the design requirements, create a sketch of initial design ideas*	100%	Graph paper, pencils	3/3/25	3/4/25	Completed
Taking Measurements - Take any necessary measurements to incorporate into our model / design concepts	100%	Measuring Tape, Pencil, Paper, Google Doc	3/8/25	3/9/25	Completed
Creating 3D models - *Create 3D models of sketches including any necessary details to show to Stakeholder*	100%	Modeling Software (Tinkercad)	3/7/25	3/9/25	Completed
Pugh Matrix (Weighted Decision Matrix) - *Determine which model / design is the one that fits the criteria that the stakeholder create*	20%	Google Docs, calculator, Weighted Decision Matrix PDF	3/11/25	3/12/25	Completed
Preliminary Engineering	----------	----------	----------	----------	
Sourcing Materials & Tools - *Communicating with Stakeholders about specific material preferences* - *Communicating with Mentors to source materials and tools* - *Communicating with Teammates about availability / access to Resources*	100%	Communicating directly with the stakeholder to find out and express their feelings towards the sourcing and availability of the possible materials.	4/4/25	3/25/25	Completed
Create a Schedule	100%	Calendar, An effective way of	2/25/25	3/25/25	Completed

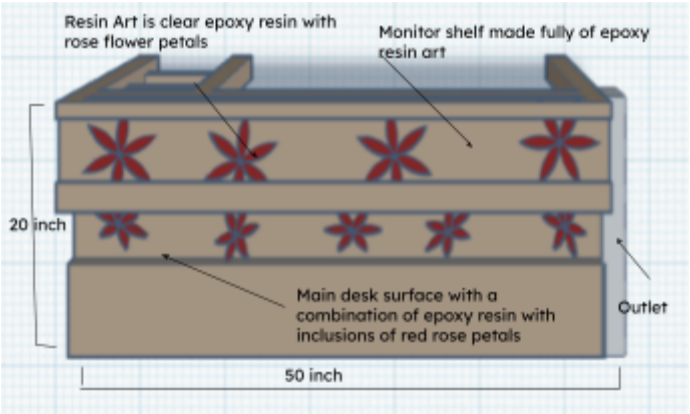
- *Communicate with team members about availability to work on project and create schedule for meet up days* - *Communicate with mentors to see availability to assist with project / offer advice*		communication with group members (Messages Group Chat), Daily email/text messages schedule of check up reminders, reviews, one on one interaction help with the project. routine/cycle to mentors			
Gathering Materials - *Making necessary purchases of materials* - *Communicate with mentors about gathering materials*	100%	Very specific material/checklist with quantities, planning to go to stores/ ordering online to get the supplies need such as the pinewood and epoxy resin.	4/18/25	4/24/25	Completed
Final Project - *Constructing final prototype* - *Communicating with mentors and teammates about schedules and possible delays*	20%	The final prototype is completed when we build all of our main compointed of the desk. The main compointed are the the resin desktop, cabinets, waterfall legs, and the montier shelf.	4/21/25	N/A	In Progress

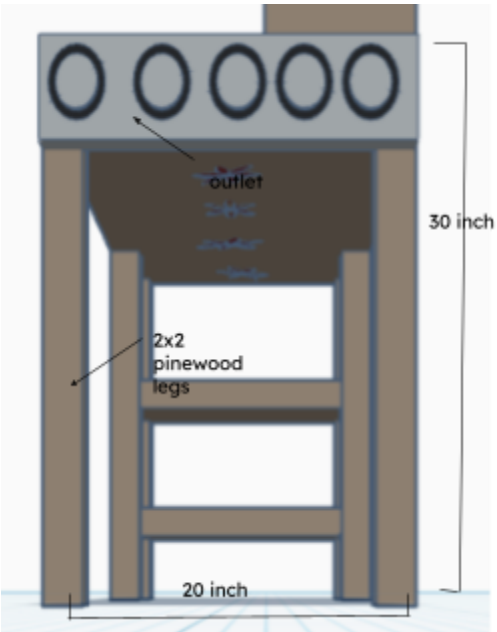
Design Blueprint

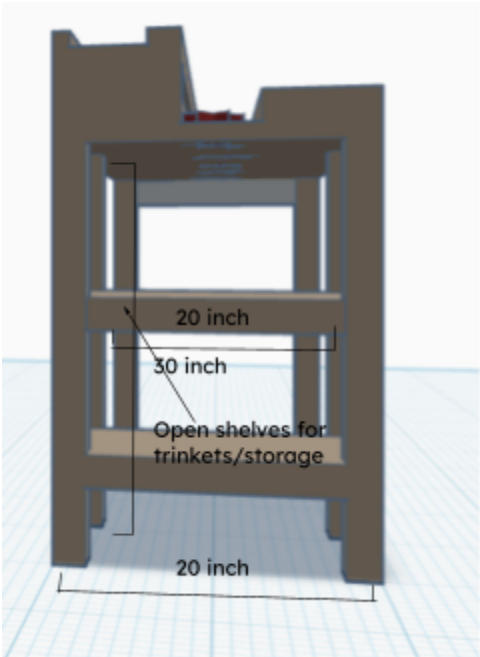
Design Blueprint #1

Date: 3/11/25
Sketch:




Resin Art is clear epoxy resin with rose flower petals
Monitor shelf made fully of epoxy resin art
20 inch
Main desk surface with a combination of epoxy resin with inclusions of red rose petals
Outlet
50 inch


outlet
30 inch
2x2 pinewood legs
20 inch



Manufacturing Methods:

Step 1: Gather all materials. Take and mark measurements on wood. Cut all necessary parts of desk out

Step 2: For the main surface area of the desk, use wood that has a gap between them to fill with resin. First pour with resin, color of choice, then use clear resin, filling it with rose petals and any other additions that the stakeholder requests. Then pour the monitor shelf. Allow to cure.

Step 3: Assemble all pieces together using nails, screws,

Supply List and Costs:

- Resin
    - $48.71 for two gallons on Amazon. A and B
    - $13.99 on Amazon for Mica Powder to color resin. 36 colors, 10 g bottles
    - Alternative to mica powder is Epoxy Resin Pigment (Also Amazon). 10 ml of 16 colors
    - *46 x 3 Resin Monitor Shelf*
    -
- Outlet extension Cord
    - Depending on Stakeholders preference for size and type of cord, price for extension cord varies.
    - Price typically starts from $9.99 for basic 6 outlet extension cord
    - Purchase might not be necessary if Stakeholder / Mentors / Teammembers have extension cord already available and approved by Stakeholder
- Wood

2x2 Table Legs| x 4 : Donated by Mr. R.N, Donated by Olivia´s father

50 x 20 Surface Table : Donated by R.N

14 x 20 Cabinet : Possibly bought used, or made from scratch

- Nails / Scews:

  $10+ from Home Depot, depending on the bulk size and brand of nails.

- Drawer railings

  $15-20 in Home Depot, depending on brand and size.